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**CAMBRIDGE INTERNATIONAL EXAMINATIONS**

**Cambridge Ordinary Level**

## MARK SCHEME for the October/November 2014 series

# 4024 MATHEMATICS (SYLLABUS D)

**4024/12** Paper 1, maximum raw mark 80

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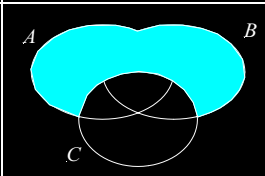
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**Abbreviations**

cao correct answer only
cso correct solution only
dep dependent
ft follow through after error
isw ignore subsequent working
oe or equivalent
SC Special Case
www without wrong working
soi seen or implied

| Question | Answers | Mark | Part marks |
|---|---|---|---|
| 1 (a) | 5.11 oe | 1 | |
| (b) | 2 hours and 35 minutes | 1 | |
| 2 (a) | 59 | 1 | |
| (b) | $T = \frac{13M}{500} + 20$ oe seen | 1 | |
| 3 (a) | –0.5 | 1 | |
| (b) | 0.1 | 1 | |
| 4 (a) | –5 | 1 | |
| (b) | $\frac{x+6}{2}$ oe | 1 | |
| 5 (a) | 1200 cao | 1 | |
| (b) | 3 | 1 | |
| 6 (a) | Correct region shaded | 1 |  |
| (b) | 3 | 1 | |
| 7 | 25 | 2 | **C1** for figs. 25<br>or **M1** for $\frac{figs\ 9}{60 \times 60}$ oe |
| 8 (a) | 1 : 2 oe | 1 | |
| (b) | 1 : 8 oe, or ft *their*(a) cubed | 1√ | |

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|---|---|---|---|---|
| **9** | **(a)** | 54.25 | **1** | |
| | **(b)** | $\frac{d+0.5}{54.25}$, or ft $\frac{d+0.5}{their(a)}$, seen | **1**√ | |
| **10** | | 12 | **2** | **B1** for "k" = 72<br>or **M1** for $9 \times 8 = 6y$ oe<br>or **M1** for $y$ = (*their k*)/6 when $y$ = "k"/$x$ used |
| **11** | **(a)** | 1 | **1** | |
| | **(b)** | 41 40 81 (all three) | **1** | |
| | **(c)** | $(2n+1)^2$ oe | **1** | |
| **12** | **(a)** | $5.67 \times 10^{-4}$ | **1** | |
| | **(b)** | $6 \times 10 -12$ | **2** | **C1** for figs 6, or for the index –12 |
| **13** | **(a)** | 140 | **1** | |
| | **(b)** | 1.2 | **2** | **M1** for $3\times\left(\frac{7}{5}-1\right)$ ; or<br>$3\times\left(\frac{their(a)}{100}-1\right)$; oe<br>or a complete algebraic method. |
| **14** | **(a)** | 10 | **1** | |
| | **(b)** | 216 | **2** | **M1** for $\pi \times 6 \times 10 = \frac{x}{360} \times \pi r^2$<br>or $2 \times \pi \times 6 = \frac{x}{360} \times 2\pi r$<br>where $r$ = 10 or *their*(a).<br>Where radians are used, method must include multiplication by $\frac{180}{\pi}$. |
| **15** | **(a)** | 720 | **1** | |
| | **(b)** | 20 | **2** | **M1** for ( $\pi \times 62 \times d$ ) (oe) = $k\pi$<br>where $k$ = 720 or *their*(a) |

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|---|---|---|---|---|
| **16** | **(a)** | $\begin{pmatrix}-4\\-3\end{pmatrix}$ | **1** | |
| | **(b)** | $\begin{pmatrix}-3\\-4\end{pmatrix}$ | **1** | |
| | **(c)** | 5 cao | **1** | |
| **17** | **(a)** | $p^5 - 3$ | **2** | **B1** for $p^5$ , or for – 3. |
| | **(b)** | $3x^2$ | **2** | **C1** for 3; **C1** for $x^2$ |
| **18** | **(a)** | $4a(1 - 4a)$ | **1** | |
| | **(b)** | $(3b - c)(3b + c)$ | **1** | |
| | **(c)** | $(x + 5)(x - y)$ | **2** | **B1** for one of the partial factorisations $x(x - y)$; $5(x - y)$; $x(x + 5)$; $y(x + 5)$, or their negatives. |
| **19** | **(a)** | 4 | **1** | |
| | **(b)** | 90° | **1** | |
| | | two 150° } correctly obtained | **1** | |
| | | two 135° } correctly obtained | **1** | If [0] earned for the two 150s, award **M1** for<br><br>using 360° correctly in a quadrilateral, or for using 540° correctly in a pentagon,<br>or for using 720° correctly in a hexagon,<br>to find the 135.<br><br>If [0] earned in (b), then **B1** for (angle sum of a hexagon equals) 720° seen. |

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|---|---|---|---|---|
| **20** | **(a)** | 68 | **1** | |
| | **(b)** | 44 | **1** | |
| | **(c)** | 112 or ft 180 – *their*(a) | **1**√ | |
| | **(d)** | 44 or ft *their*(b) | **1**√ | |
| **21** | **(a)** | Correct completion of tree diagram | **1** | |
| | **(b)** | **(i)** $\frac{1}{10}$ | **1** | |
| | | **(ii)** $\frac{17}{50}$ or ft from *their* tree diagram | **2**√ | **M1** for $\left\{\frac{2}{5}\times\frac{1}{4}\text{ or }their(bi)\right\}+\frac{3}{5}\times their\left(\frac{2}{5}\right)$ |
| **22** | **(a)** | 1.2 | **1** | |
| | **(b)** | 3.6 | **1** | |
| | **(c)** | 480 | **2** | **M1** for $\frac{1}{2}\times(20+60)\times 12$ oe<br>or **B1** for 180, or 240, or 60, or 420, or 300, as a correct evaluation of an identifiable appropriate area. |
| **23** | **(a)** | (8, 10) | **1** | |
| | **(b)** | $x > 8$ oe<br>$2y > 12 + x$ oe | **1**<br>**1** | If 0 scored, then **C1** for $x \geqslant 8$ oe **and** $2y \geqslant 12 + x$ oe. |
| | **(c)** | (9, 11) | **1** | |
| **24** | **(a)** | 137° to 140° inclusive | **1** | |
| | **(b)** | **(i)** perp. bisector of *AB* | **1** | |
| | | **(ii)** circle, centre *C*, radius 4 cm | **1** | |
| | | **(iii)** correct region (bottom part) shaded | **1** | |

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|---|---|---|---|---|
| **25** | **(a)** | $\left(-\frac{1}{2}, 1\right)$ | **1** | |
| | **(b)** | $-\frac{6}{7}$ | **1** | |
| | **(c)** | **(i)** (10, –8) | **2** | **C1** for one correct coordinate |
| | | **(ii)** $\frac{1}{3}$ | **1** | |
| **26** | **(a)** | $\frac{1}{7}$ | **1** | |
| | **(b)** | $\begin{pmatrix} -1 & -4 \\ 2 & 0 \end{pmatrix}$ | **2** | **C1** for 2 or 3 correct elements. |
| | **(c)** | (2  0), or (14 × *their* (a)  0) ft | **2**√ | M1 for ( **Y** =)  ( 6  2 ) $\mathbf{A}^{-1}$ seen.<br>If ($x$  $y$) **A** = (6  2) is used, then award **M1**<br>at the stage where an attempt to solve the simultaneous eqns. is made. |

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